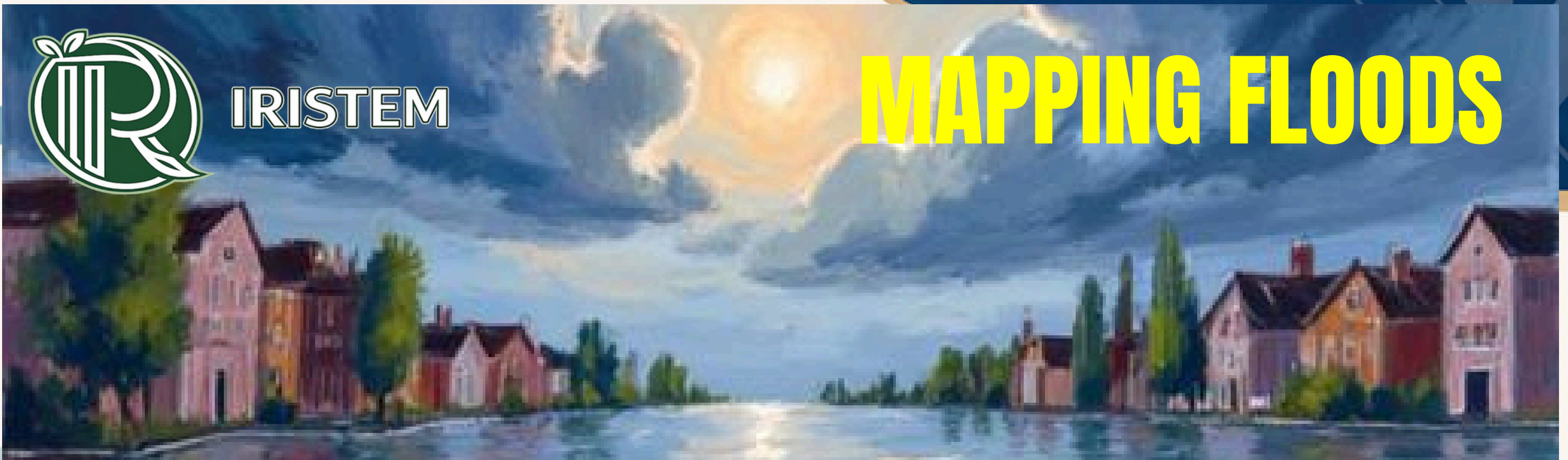




IRISTEM

MAPPING FLOODS



Abstract

Flooding can be dangerous for coastal communities during strong storms and heavy rain. In **Coulibistrie, Dominica**, researchers and residents worked **together to understand** which places might flood. Researchers used **drone pictures** and maps to study the community from above. Residents added what they knew

from past storms, such as which roads flooded first and which areas were harder to reach. **Maps, technology, and community knowledge** can help people prepare before the next disaster.

Introduction

Coulibistrie is a coastal community in Dominica that can be affected by storms, **heavy rainfall**, and flooding. Flooding does not affect every place the same way. Some roads may flood quickly, while other areas may be safer because they are higher up or farther from water. The project asked an important question: How can maps help a community **understand** flood risk before a disaster happens?



Small island communities face growing climate risks and need ways to prepare for future disasters.

How do Scientists Map Risk

Scientists and residents used maps to better understand **flood risk** in Coulibistrie. Researchers used **drone pictures** to see buildings, roads, rivers, and coastlines from above. Residents then **marked important places** on maps, such as homes, bridges, safe areas, and places that had flooded before. This helped combine scientific tools with local knowledge, since people who live in the community often know what maps alone might miss.



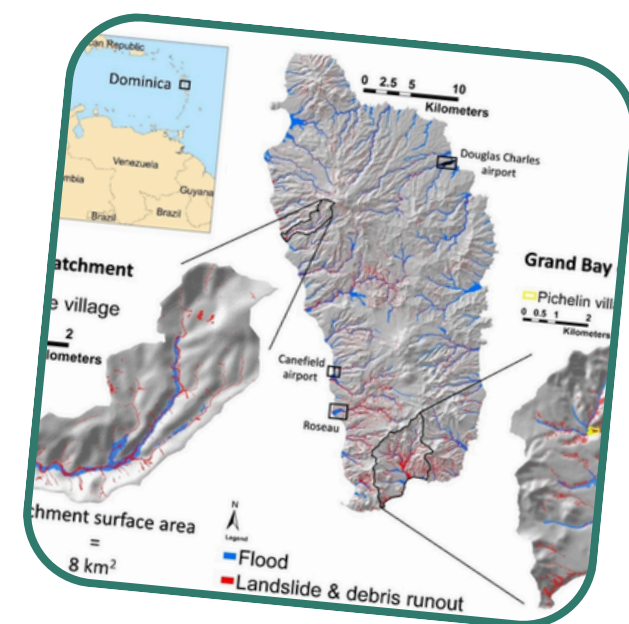
1. Look From Above

Drone pictures help scientists see roads, rivers, homes, and coastlines from a bird's-eye view.



2. Listen to Residents

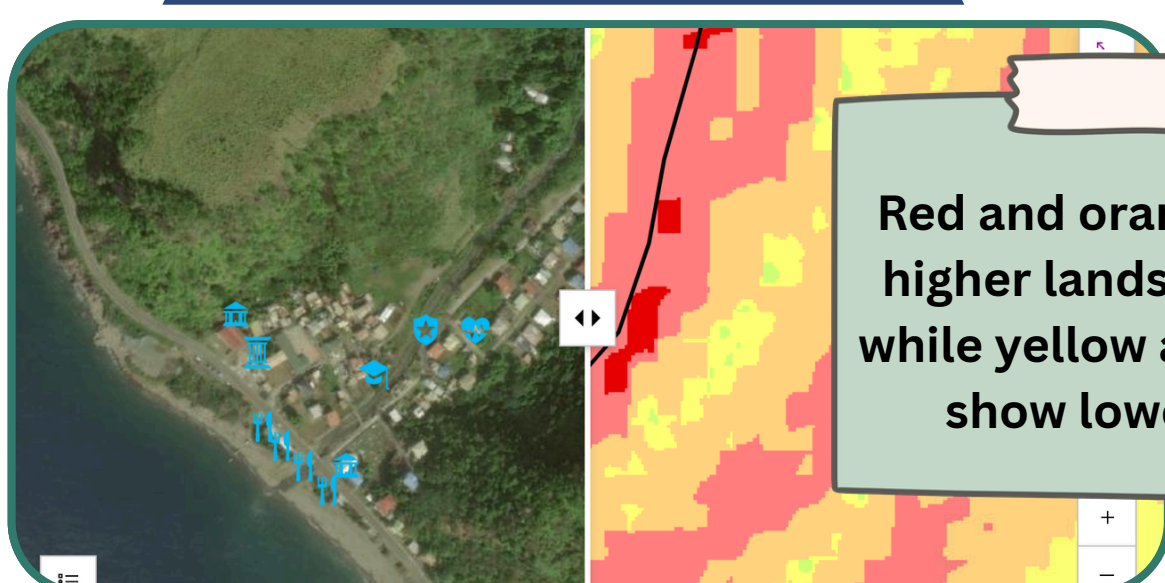
People who live in the community know which places flooded before and which roads may become unsafe.



3. Make a Risk Map

Scientists and residents combine their information to mark risky areas, safer places, and possible evacuation routes.

LANDSLIDE



Red and orange show higher landslide risk, while yellow and green show lower risk

FLOOD



Red and orange show higher flood risk, while yellow and grey show lower risk

What did they Discover

The project showed that maps are more useful when they include both science and local knowledge. **Drone pictures** can show where roads, rivers, homes, and coastlines are located, but **residents can explain** how those places are affected during real storms. A road may look safe on a map, but people nearby may know it floods after heavy rain. This shows why disaster planning should include the people who **know the community best.**



Flood risk matters because it affects real homes, roads, and community spaces.

Takeaways

Maps Show Risk

Maps can help people see which places may flood or become unsafe.

Technology helps

Drone pictures let people study a community from above.

Local knowledge matters

Residents often know details that maps alone cannot show.

Planning saves time

Communities can use maps to prepare before a disaster happens.

Key Terms

Flood — When water covers land that is usually dry.

Flood Risk — The chance that a place could be damaged by flooding.

Drone — A small flying machine that can take pictures from above.

Drone Imagery — Pictures taken by a drone.

Community Mapping — When people work together to mark important places, risks, and safe areas on a map.

Hazard — Something that could cause danger or damage.

Climate Resilience — A community's ability to prepare for and recover from climate problems.

Evacuation Route — A path people use to leave an unsafe area.

Flood Ready Challenge

- Draw a small coastal town with a river, roads, homes, a school, and high ground.
- Use blue to mark places that might flood.
- Use red to mark dangerous places.
- Use green to mark safe places.
- Draw arrows to show where people should go.

Your group has 5 planning points to help the town prepare.

- 1 point — Warning signs
- 1 point — Evacuation route
- 2 points — Safe meeting place
- 2 points — Protect a bridge or road
- 3 points — Move people or supplies to high ground

Afterward, explain what your group protected first and why.

References

ArcGIS StoryMap. Coulibistrie, Dominica.

<https://storymaps.arcgis.com/stories/1b49c1a74cab42b8b7897044cf8501d2>